

Khulna University of Engineering & Technology  
Department of Building Engineering and Construction Management  
B. Sc. Engineering 2<sup>nd</sup> Year 1<sup>st</sup> Term Regular Examination, 2018  
**Math 2123**  
(Mathematics III)

Full Marks: 210

Time: 3 hrs

- N.B. i) Answer any three questions from each section in separate script.  
ii) Figures in the right margin indicate full marks.

**Section - A**

1. (a) Define orthogonal matrix, Hermitian matrix, Idempotent matrix and lower triangular matrix with example. (12)

- (b) Express the matrix  $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ 2 & -1 & -3 \end{bmatrix}$  as the sum of a symmetric and skew-symmetric matrix. (10)

- (c) Find the inverse of the following matrix applying elementary row transformation. (13)

$$A = \begin{bmatrix} -2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \times (-1)} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + R_3} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A^{-1}$$

2. (a) Reduce the matrix  $A = \begin{bmatrix} 2 & 1 & -1 & 3 \\ -3 & 0 & 1 & -2 \\ -4 & 5 & 0 & -1 \end{bmatrix}$  to echelon form and then to canonical form and then to normal form. Hence find its rank. (12)

- (b) Find the non-singular matrices P and Q such that PAQ is the normal form, where  $A = \begin{bmatrix} 2 & -1 & 3 & 4 \\ -1 & -3 & 0 & 1 \\ 1 & -2 & -1 & 2 \end{bmatrix}$ . (15)

- (c) Show that the diagonal elements of a skew-symmetric matrix are zero. (08)

3. (a) When the system of linear equations are called consistent? Determine the value of a so that the following equations has (i) no solution (ii) non-zero unique solution (iii) Infinite solution: (14)

$$x+y-z=1; 2x+3y+az=3; x+ay+3z=2.$$

- (b) Check the following system of linear equations are consistent or not. If consistent then find the non-trivial solutions of them: (11)

$$2x_1 - 3x_2 + x_3 + 5x_4 = 2$$

$$-x_1 - 2x_3 + x_4 = 3$$

$$3x_1 + 2x_2 - x_3 + 2x_4 = -1$$

4. (a) Define elementary matrix and equivalent matrix with examples. (07)

- (b) Show that if A is involutory then  $\frac{1}{2}(I+A)$  and  $\frac{1}{2}(I-A)$  are Idempotent. (08)

$$A \text{ is involutory so, } A^2 = I$$

$$B = \frac{1}{2}(I+A) \quad C = \frac{1}{2}(I-A)$$

$$B^2 = \frac{1}{4}(I+2A+I) = \frac{1}{4}(I-2A+I) = \frac{1}{4}(2I) = \frac{1}{2}I = B$$

$$= \frac{1}{2}(2I+2A) = \frac{1}{2}(2I) = I = B$$

$$= \frac{1}{2}(I+I) = I = B$$

$$\therefore B, C \text{ both idempotent}$$

- (c) Define Eigen values and Eigen vector and write its three properties. And find the Eigen values and Eigen vector corresponding to one of the Eigen values of the matrix  $B = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$

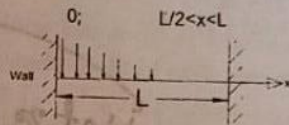
### Section - B

5. (a) State the sufficient conditions for the existence of Laplace transform of functions. Find Laplace transform of  $f(t) = \frac{1}{\sqrt{t}}$ ,  $0 < t < \infty$  if exist. (12)
- (b) Find the Laplace transform of  $g(t)$ , where  $g(t) = \int_0^t \frac{\sin u}{u} du + te^{2t} \cos 2t$ . (13)
- (c) Using convolution theorem find the inverse Laplace transform of  $\frac{1}{s^2(s+1)^2}$ . (10)

6. (a) Using Laplace transform solve the following equation (15)

$$\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = e^t \text{ with } x = 2, \frac{dx}{dt} = -1 \text{ at } t = 0$$

- (b) A beam of length  $L$  is embedded at both ends as shown in figure below. Find the deflection of the beam when the load is given by  $w(x) = w_0(1 - \frac{x}{L})$ ;  $0 < x < \frac{L}{2}$  (20)



7. (a) Geometrically interpret the scalar triple product and use it to determine whether the following set of vectors are linearly dependent or not: (10)

$$\vec{A} = \vec{i} - 3\vec{j} + 2\vec{k}, \vec{B} = 3\vec{i} + 2\vec{j} - \vec{k}, \vec{C} = 2\vec{i} + \vec{j} - 3\vec{k}$$

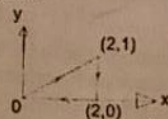
- (b) A force  $\vec{F} = 3\vec{i} + 2\vec{j} - 4\vec{k}$  is applied at the point  $(1, -1, 2)$ . Find the moment of the force about the point  $(2, -1, 3)$ . (10)

- (c) Show that the vector field represented by

$$\vec{F} = (z^2 + 2x + 3y)\vec{i} + (3x + 2y + z)\vec{j} + (y + 2zx)\vec{k} \text{ is not solenoidal. Is } \vec{F} \text{ conservative? If so, find its scalar potential } \phi.$$

- (a) Find the value of  $a$  and  $b$  so that the surface  $ax^2 - byz = (a+z)$  will be orthogonal to the surface  $4x^2y + z^2 = 4$  at the point  $(1, -1, 2)$ . (10)

- (b) If  $\vec{F} = (2x + y^2)\vec{i} + (3y - 4x)\vec{j}$  evaluate the  $\int_C \vec{F} \cdot d\vec{r}$  around the triangle  $c$  of below figure in the directed path: (10)



- (c) Evaluate  $\iint_S \vec{A} \cdot \vec{n} ds$ , where  $\vec{A} = z\vec{i} + x\vec{j} - 3y^2z\vec{k}$  and  $s$  is the surface of the cylinder  $x^2 + y^2 = 16$  included in the first octant between  $z=0$  and  $z=5$ . (15)